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Human papilloma virus (HPV) is the main cause of cervical cancer. A vaccine has been developed to protect girls and women from HPV.

Describe how giving this vaccine leads to production of antibody against HPV.

[4 marks]



07 . 2

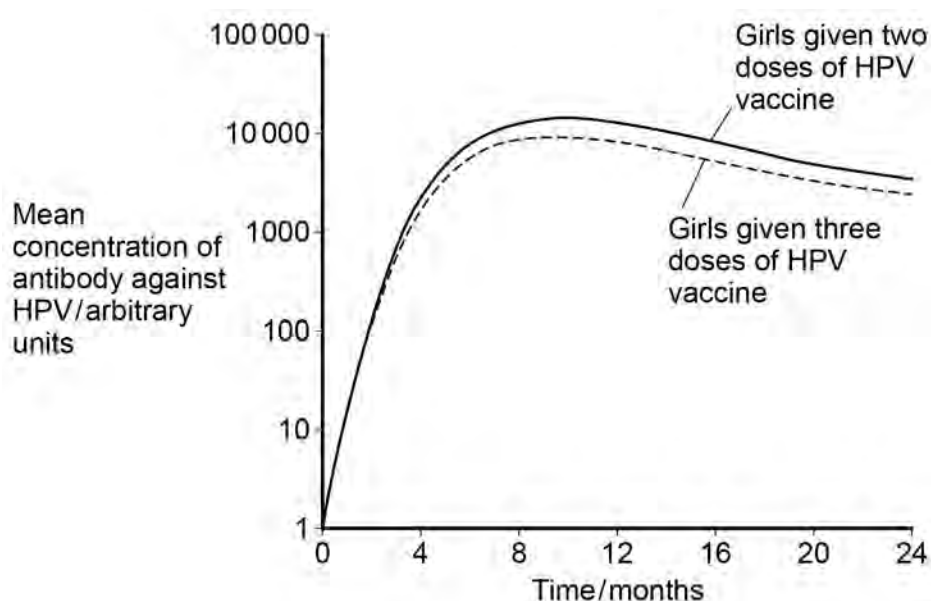
Doctors investigated whether it was better to give two or three doses of the HPV vaccine. They determined the mean concentration of antibody against HPV in blood samples from girls who were given either two or three doses of the vaccine.

- Girls given two doses received an initial vaccination, followed by a second at 6 months.
- Girls given three doses received an initial vaccination, followed by a second at 1 month and a third at 6 months.

The doctors measured the concentration of antibody each month.

The results are shown in **Figure 5**.

Figure 5



What do these results suggest about whether it is better to give two or three doses of the vaccine? Give reasons for your answer.

[2 marks]

Question 7 continues on the next page



0 7 . 3

The doctors carried out a statistical test to determine whether the antibody concentrations were significantly different in girls given two doses of the vaccine, compared with those given three doses. They determined the mean concentrations of antibody 9 months after the first dose of vaccine.

What statistical test should the doctors have used? Give the reason for your choice.

[1 mark]

Test _____

Reason _____

0 7 . 4

There is genetic diversity within HPV.

Give **two** ways doctors could use base sequences to compare different types of HPV.

[2 marks]

1 _____

2 _____

